Minsu Kwak

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RESEARCH INTERESTS

Neutrino physics · event reconstruction · detector simulation · machine learning for physics

EDUCATION

2026 – present **M.S.–Ph.D. Integrated Program in Physics**, Chung-Ang University
Particle physics. Advisor: Prof. Changhyon Ha

2020 – 2026 **B.S. in Physics**, Dankook University

RESEARCH EXPERIENCE

2025 – present **Rare Event Search Lab**, Chung-Ang University (IceCube Collaboration)
Undergraduate researcher from August 2025; graduate researcher from March 2026

Jun – Aug 2026 **Visiting researcher**, RWTH Aachen University, Germany
High-energy event reconstruction for IceCube

RESEARCH PROJECTS

IceCube Upgrade **Per-PMT photon tables and Monopod reconstruction**
Photon tables for each PMT of the multi-PMT Upgrade modules (mDOM, D-Egg): CLSim tabulation on HTCondor, photospline fits and run-time ice anisotropy, used in Millipede/Monopod cascade reconstruction of GeV neutrinos.

CANDY **Liquid-scintillator detector with an in-liquid SiPM array**
Geant4 simulation validated against measured photon counts; neural-network reconstruction of muon tracks and positron vertices; reactor sterile-neutrino sensitivity.

PUBLICATIONS

2026 H. Kimku, J. S. Chung, C. Ha, T. Z. Huang, J. Kim, S. Kim, B. C. Koh, **M. S. Kwak**, S. Lee, Y. J. Lee, J. Seo, *Charged-particle topology reconstruction with an in-liquid SiPM array*, arXiv:2609.25532.

SOFTWARE

2026 **IceTray** (IceCube software framework), pull request #4621: per-PMT photonics tables for multi-PMT modules (mDOM, D-Egg) in clsim, photonics-service and millipede.

PRESENTATIONS

Oct 2026 **IceCube Fall Collaboration Meeting**, Madison, Wisconsin, Oct 19–23 (talk, upcoming)
Developing likelihood-based cascade reconstruction for the Upgrade

Oct 2026 **K-Neutrino Symposium 2026**, NNN26 satellite program (poster, upcoming)
Per-PMT spline tables for GeV neutrino reconstruction in the IceCube Upgrade

Oct 2026 **NNN26**, 25th International Workshop on Next Generation Nucleon Decay and Neutrino Detectors, Chung-Ang University, Seoul (poster, upcoming)
Charged-particle topology reconstruction with an in-liquid SiPM array

May 2026 **IceCube Spring Collaboration Meeting**, Bochum, Germany, May 18–22 (talk)
Monopod systematic study

Oct 2025	**KPS Fall Meeting**, Korean Physical Society (poster) Performance evaluation of the LS-based CANDY detector using Geant4 simulation and CNN-based reconstruction
Oct 2025	**CPNR Scientific Program 2025** (talk) CANDY's resolution and positioning study using Geant4

HONORS

2026	Best Presentation Award, KPS Spring Meeting, Korean Physical Society
2020 – 2025	Academic Excellence Scholarship, Dankook University (seven semesters)

SKILLS

Programming	C++, Python, Bash
Physics software	IceTray (CLSim, photonics-service, Millipede), photospline, Geant4, ROOT
Computing	HTCondor, Linux clusters, PyTorch
Methods	Monte Carlo simulation, maximum-likelihood reconstruction, CNNs, particle swarm optimization